

Understanding the Limits of Single-Cell Foundation Models on Downstream Tasks

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Motivation and objective. Foundation models based on transformer architectures have recently emerged as a powerful paradigm for single-cell transcriptomics. Geneformer demonstrated that large-scale pretraining on gene expression sequences enables transferable representations for diverse downstream tasks [1]. scGPT further introduced a generative pretraining objective and showed strong performance across multiple single-cell applications [2]. More recently, scFoundation scaled training to tens of millions of human transcriptomes and reported competitive results across benchmarks [3]. Despite these advances, it remains unclear what biological information is robustly encoded during pretraining and under which conditions these representations fail to generalize. The objective of this project is to provide a controlled and interpretable evaluation of representative single-cell foundation models, with a particular focus on their limitations.

Problem formulation. Reported performance of single-cell foundation models is often difficult to compare due to heterogeneous datasets, preprocessing pipelines, and fine-tuning strategies. This project studies downstream cell type classification under a strictly unified experimental protocol in order to isolate the contribution of pretraining. We hypothesize that foundation models provide strong gains when downstream data are aligned with the pretraining distribution, while their advantage diminishes under domain shift, where representation quality rather than task-specific optimization becomes decisive.

Datasets. The primary evaluation will be conducted on the 10x Genomics PBMC 68k dataset, a widely used peripheral blood mononuclear cell benchmark with curated cell-type annotations [4]. This dataset enables reproducible comparison with prior work and provides a well-understood baseline setting. To assess generalization beyond the pretraining distribution, a secondary evaluation will be performed on the Tabula Sapiens human cell atlas, which comprises single-cell transcriptomes from multiple organs and donors [5]. This cross-dataset setting explicitly probes robustness and domain-shift behavior.

Method overview. All models will be evaluated using identical preprocessing, gene selection, and train/validation/test splits. For each foundation model, cell embeddings will be extracted and used for downstream classification. Two regimes will be compared: frozen representations with a lightweight classifier and task-head fine-tuning under identical optimization settings. This design separates the effect of representation quality from downstream optimization.

Evaluation and analysis. Performance will be quantified using accuracy and macro-F1 score, with stratified analysis by cell-type frequency to assess robustness to class imbalance. Embedding quality will be further examined through low-dimensional visualization and confusion analysis to identify systematic failure modes. The cross-dataset evaluation from PBMC to Tabula Sapiens will be used to explicitly test generalization beyond the training distribution.

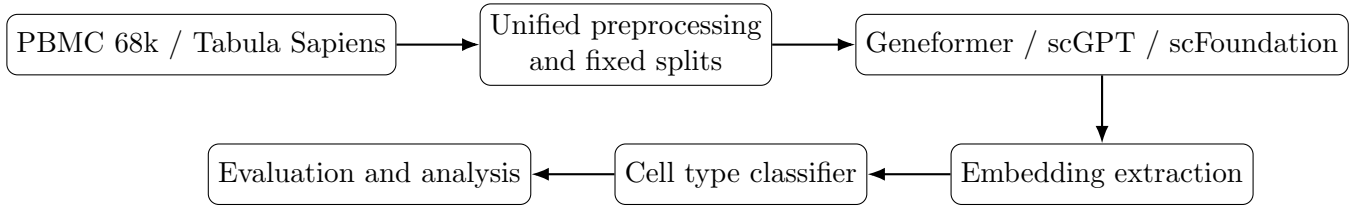


Figure 1: Unified evaluation pipeline for single-cell foundation models. All components except the pretrained encoder are kept identical across experiments.

Expected contribution. This project will deliver a reproducible and interpretable comparison of representative single-cell foundation models. By focusing on standardized evaluation, frozen versus fine-tuned usage, and domain-shift analysis, the study aims to clarify when pretraining provides tangible benefits and where current models fall short. The results are expected to inform both practical model selection and future methodological development.

External support and AI tools. Any external support or AI-based tools used for writing or coding will be explicitly disclosed in the final report, in accordance with course policy.

References

- [1] C. V. Theodoris et al. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023. doi:10.1038/s41586-023-06139-9.
- [2] H. Cui et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480, 2024. doi:10.1038/s41592-024-02201-0.
- [3] M. Hao et al. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 2024. doi:10.1038/s41592-024-02305-7.
- [4] G. X. Y. Zheng et al. Massively parallel digital transcriptional profiling of single cells. *Nature Communications*, 8:14049, 2017. doi:10.1038/ncomms14049.
- [5] Tabula Sapiens Consortium. The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896, 2022. doi:10.1126/science.abl4896.